

Course outline

SFWRENG 2AA4 Software Design I: Introduction to Software Development (2025 Winter)

INSTRUCTORS

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 - Office hours: by appointment
- **Teaching assistants**
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GENERAL INFORMATION

CALENDAR DESCRIPTION

Software life cycle, quality attributes, requirements documentation, specifying behavior; classes and objects, interface specification; creational, structural, and behavioral software design patterns; implementation in code, reviews, testing and verification.

LEARNING OBJECTIVES¹

This course introduces students to the profession of software engineering and the software development process. Students will gain an appreciation of software design strategies that facilitate working on large applications with multiple developers. Students will study the basic principles of software design. They will learn how to specify, design, implement, and verify software units in an object-oriented language like Java. It builds on the knowledge acquired in **SFWRENG 2DM3** (*Discrete Mathematics with Applications I*), **SFWRENG 2OP3** (*Object-oriented Programming*) and the experiential one from **SFWRENG 2XC3** (*Software Engineering Practice and Experience: Development Basics*). At the end of the course, students should:

- **Know and understand:**
 - The engineering dimension associated with developing software.
 - The principles guiding the specification, design and implementation of software.
 - The importance of incremental deliveries and iterative/collaborative development.
- **Be able to:**
 - Specify software at large-scale and small scale
 - Use state-of-practice tools to implement software collaboratively
 - Design (object-oriented) code according to engineering principles
 - Deliver code in a professional way

¹ Based on the descriptions of Dr. Spencer Smith's (2AA4 – Winter 2020) and Dr. Sebastien Mosser (2AA4 – Winter 2024).

COURSE PAGE

The course uses *Avenue2Learn* for pedagogical material.

SCHEDULE

Slot	Monday	Tuesday	Wednesday	Thursday	Friday
10:30AM 11:20 AM	Lecture		Lecture	Lecture	T02
11:30 AM 12:20 PM					
12:30 PM 1:20 PM	T01				
1:30 PM 2:20 PM					
2:30 PM 3:20 PM		T03			
3:30 PM 4:20 PM					

Attendance of lectures and tutorial is not mandatory but highly recommended.

Instructors reserve the right to cancel a scheduled lecture according to unexpected circumstances.

TEXTBOOK AND COURSE MATERIAL

No textbook is mandatory for this course. Slides from the lecture will be available on Avenue (but are not a textbook or lecture notes). The following books are “recommended” books that complement the course material. The bibliography was curated to contain reference books from a state-of-practice point of view, freely available for students through McMaster Library:

- Introduction to software design in Java (Martin Robillard, 2022)
 - Freely available as PDF in [SpringerLink](#) for McMaster's student
- Clean code (Robert Martin, 2008)
 - Freely available online through [Safari](#) for McMaster's student
- Software Engineering at Google (Winters et al, 2020)

ONLINE COMPONENT & COMMUNICATION

The course uses Avenue as its main website to make lecture slides, tutorial material and assignments available. To store code, we will use the GitHub platform so that students can start to build their portfolio.

COURSE CONTENTS

OUTLINE

The course will (tentatively) cover the following topics.

#Week	Lecture date	Topic	Format
1	Jan 6, 2025	Welcome to 2AA4	In person
	Jan 8, 2025	Software Development & Software Engineering 1	In person
	Jan 9, 2025	Software Development & Software Engineering 2	In person
2	Jan 13, 2025	Iterative and incremental development	In person
	Jan 15, 2025	Code quality	In person
	Jan 16, 2025	Lifecycle models	In person
3	Jan 20, 2025	UML 1	In person
	Jan 22, 2025	UML 2	In person
	Jan 23, 2025	Encapsulation	In person
4	Jan 27, 2025	SOLID principles	In person
	Jan 29, 2025	Modular design	In person
	Jan 30, 2025	Types and interfaces	In person
5	Feb 3, 2025	Comparing things	In person
	Feb 5, 2025	Composition	In person
	Feb 6, 2025	Inheritance and subtyping	In person
6	Feb 10, 2025	Testing	In person
	Feb 12, 2025	Unicity and equivalence	In person
	Feb 13, 2025	MIDTERM	
7	Recess week – No lectures		
8	Feb 24, 2025	Midterm review + wrap up Part I	In person
	Feb 26, 2025	Design patterns: Intro	In person
	Feb 27, 2025	Design patterns (C): Singleton, Factory	Video
9	Mar 3, 2025	Principles: ISP, Overloading + DP (C) discussion	In person
	Mar 5, 2025	Absence of value	In person
	Mar 6, 2025	Design patterns (B): Iterator, Observer (+a bit of MVC), Visitor	Video
10	Mar 10, 2025	Principles: Equality, Liskov, IoC, DI + DP (B) discussion	In person
	Mar 12, 2025	Design patterns (S1): Flyweight, Adapter, Decorator	Video
	Mar 13, 2025	STUPID	In person
11	Mar 17, 2025	Design patterns (S2): Command, Chain of responsibility, Template method, Closing remarks	Video
	Mar 19, 2025	Design patterns: wrap-up	In person
	Mar 20, 2025	Advanced patterns (MVP + Fluent API)	In person
12	Mar 24, 2025	Duck strategies	In person
	Mar 26, 2025	MIS and ADT	In person
	Mar 27, 2025	Architectural design	In person
13	Mar 31, 2025	Documentation	In person
	Apr 2, 2025	Evolution	In person
	Apr 3, 2025	Requirements engineering	In person
14	Apr 7, 2025	Wrap-up	In person

The instructor reserves the right to modify this tentative planning during the term. If any modification becomes necessary, reasonable notice and communication with the students will be given with an explanation and the opportunity to comment on changes.

GRADING & EVALUATION

ID	Type	Kind	Weight	Available	Due
Assignment 1	Assignment	Individual	10%	Jan 13, 2025	Feb 3, 2025
Assignment 2	Assignment	Group	15%	Feb 3, 2025	Mar 17, 2025
Midterm test	Assessment	Individual	25%	Feb 13, 2025	
Assignment 3	Assignment	Individual	10%	Mar 17, 2025	Mar 31, 2025
Final exam	Assessment	Individual	40%	TBA (Registrar)	

Assignments reports are delivered on Avenue. Source code is delivered on GitHub.

Group assignments are expected to be executed by groups of three students **from the same tutorial section**. Groups of two students will be considered if the number of students does not divide by three.

Deliveries are expected at 23:59:59 ET of the due date. Late submissions are not graded.

The instructor reserves the right to modify this planning during the term. If any modification becomes necessary, reasonable notice and communication with the students will be given with an explanation and the opportunity to comment on changes.

MSAF POLICY

Type-A MSAFs are automatically accepted when submitted with the following relief:

- For an **assignment**, it gives the student submitting the request **three (3) additional days** to finish their part in the project.
- For the **midterm**, the exam is cancelled, and **the weight carries on to the final** (now weighted as 50% of the final grade).
- **There is no need to contact the instructor. Submit your MSAF through Mosaic.**

Type-B MSAFs will be discussed with the student after being accepted by the Faculty.

ACADEMIC INTEGRITY & PLAGIARISM²

You are expected to exhibit honesty and ethical behavior in all aspects of the learning process. The academic credentials you earn are rooted in honesty and academic integrity principles. Individual assignments must be solved by one person only, **any** outside source (this includes asking other people or using any books or information found on the web) **has** to be documented. In compliance with the senate regulations on academic integrity, I remind you that: **People who let other people copy are as guilty as the ones who copy**. You can consult outside sources, meaning textbooks or the web, but **any** use of an outside source **must** be documented. **Code plagiarism detection tools will be used systematically.**

² Adapted from Dr. Jacques Carette outline for SFWRENG 3GB3 – Fall 2021

Similarly, group assignments must be solved by members of that group only, and the University policies apply as well (see “Advisory Statements” below).

MAPPING TO GRADUATE ATTRIBUTES

Attribute	Indicator		Evaluation	
			A _i	Ex.
Knowledge Base	1.1	Competence in Mathematics	✓	
	1.4	Competence in specialized engineering knowledge	✓	✓
Problem analysis	2.1	Identifies and states reasonable assumptions and suitable engineering fundamentals, before proposing a solution path to a problem	✓	✓
Investigation	3.1	Selects appropriately from relevant knowledge base to plan appropriate data collection methods and analysis strategies	✓	
	3.2	Synthesizes the results of an investigation to reach valid conclusions	✓	
Design	4.1	Defines the problem by identifying relevant context, constraints, and prior approaches before exploring potential design solutions	✓	✓
	4.2	Explores a breadth of potential solutions, considering their benefits and trade-offs as they relate to the project requirements	✓	✓
	4.3	Develops models/prototypes; tests, evaluates, and iterates as appropriate	✓	
Use of engineering tools	5.2	Successfully uses engineering tools	✓	
Professionalism	8.1	Describes the duty of a Professional Engineer to the public, client, employer, and the profession	✓	✓

ADVISORY STATEMENTS

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the **Academic Integrity Policy**, located at <https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE COMPONENT

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect,

disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities policy](#).

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors. The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.